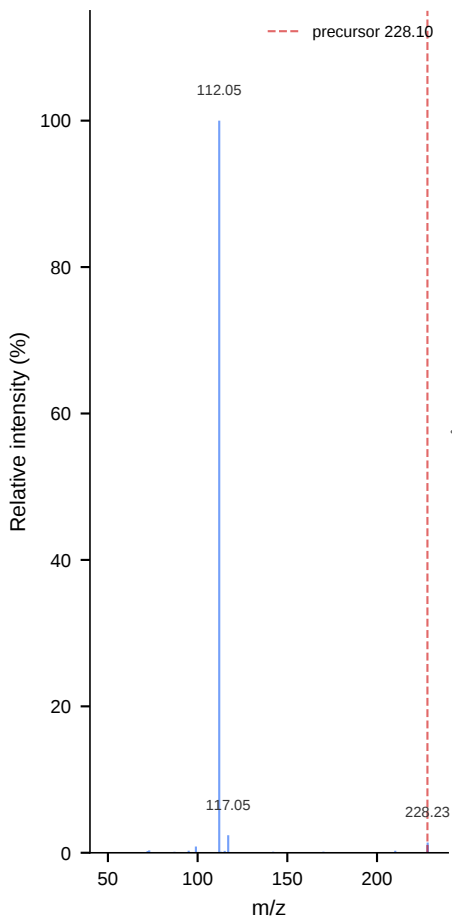


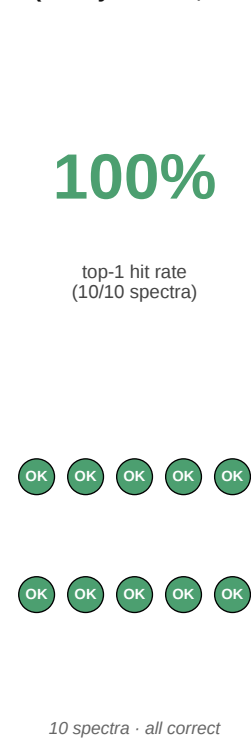
End-to-end case study from raw spectra to pathway identification

Task e2e_enrich_mammalian_RAMP_P_000050099_seed1975252413 · GT pathway: Pyrimidine catabolism (reactome · R-HSA-73621)

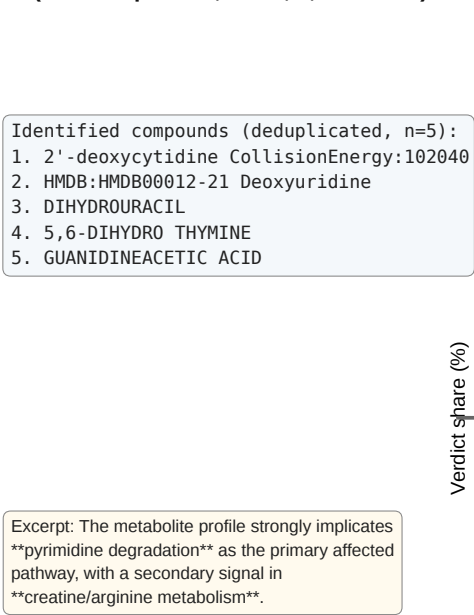
a Input: 1 of 10 MS/MS spectra
positive mode · [M+H]¹⁺ · 37 peaks



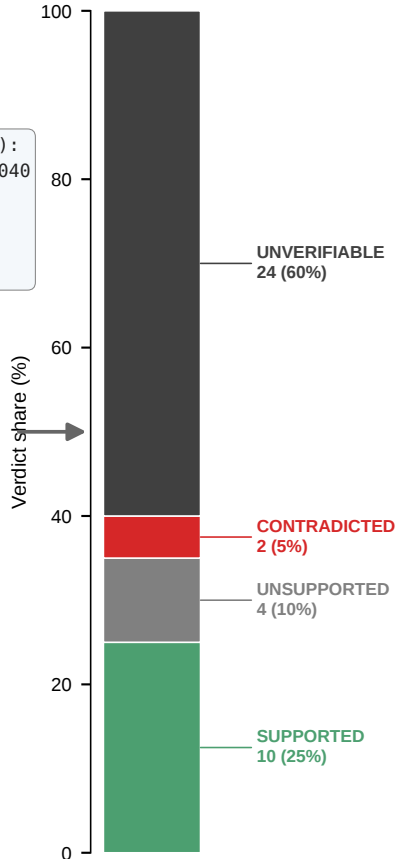
b Stage 1 — Library ID
(library_search, 1.2s)



c Stage 2 — LLM narrative
(claude-opus-4-7, 17.3s, 2,738 chars)



d Stage 3 — Verifier
(40 extracted claims)



e Final — Pathway identification

